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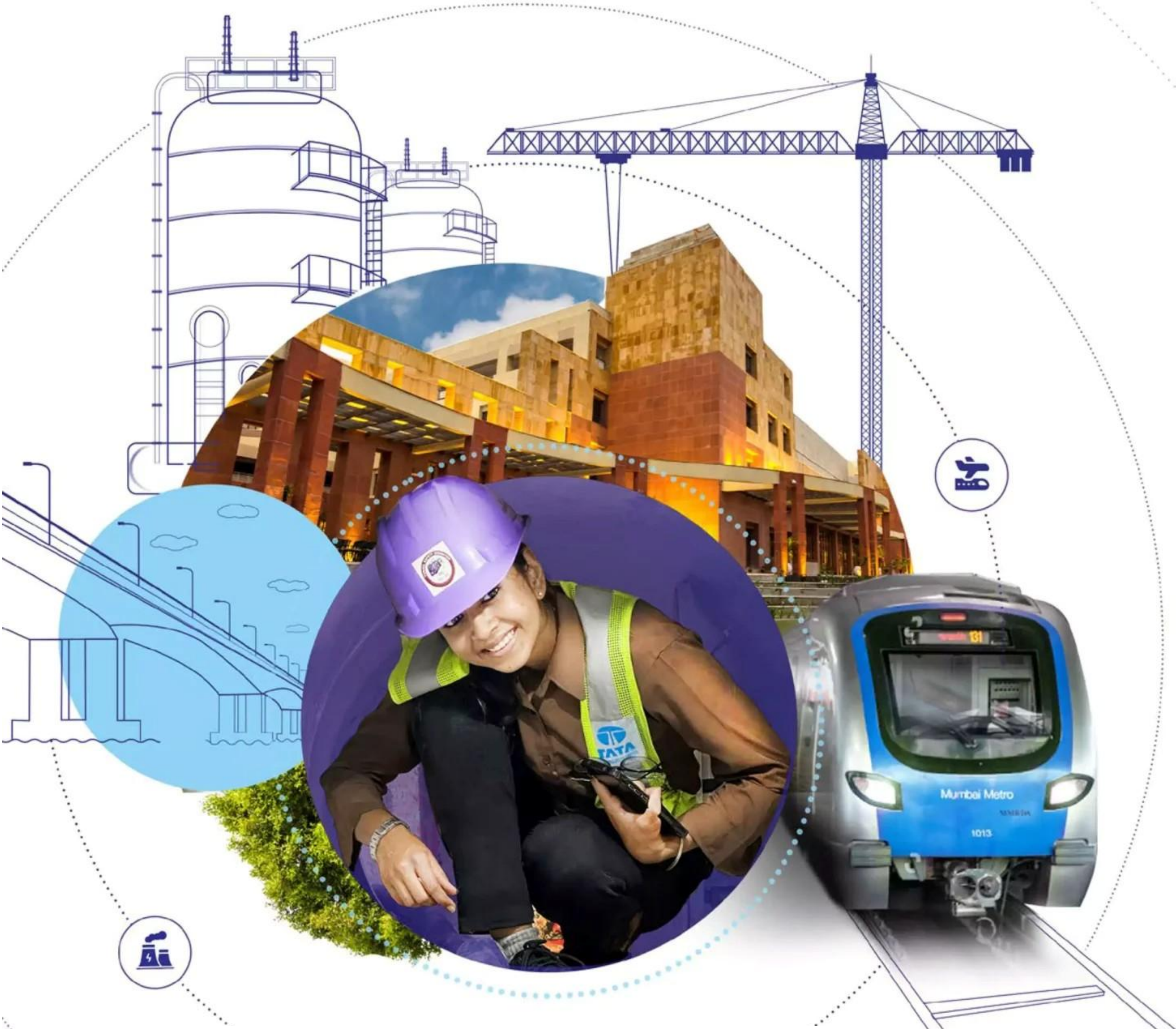

Vocational Training Report

Submitted by: Komal Baraskar

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ACKNOWLEDGEMENT

This vocational training carried out by me has been possible only because of the exceptional help, support, and guidance from **TATA Projects Ltd., TMU**.

I would like to extend my special thanks to **Mr. Sachin Gupta** *for giving me the valuable opportunity to undertake this vocational training.*

I express my sincere gratitude to **Mr. Ravi Mahule**, **Mr. Sameer Baveja** & **Mr. Satpal Singh** *for giving me the opportunity to undertake this vocational training and for their valuable support throughout.*

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Finally, I extend my sincere appreciation to **TATA Projects Ltd., TMU** *for providing me with this valuable opportunity and exposure to industrial practices.*

KOMAL BARASKAR

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➢ INTRODUCTION

Vocational training is an essential part of engineering education that bridges the gap between academic knowledge and industrial practice. It provides students with an opportunity to understand real-world manufacturing processes, workplace safety, quality standards, and industrial management systems.

I completed a 30-day vocational training at **Tata Projects Ltd.**, Structural Fabrication Division, where I gained practical exposure to the complete fabrication workflow from raw material inspection to final dispatch of fabricated structures. During this training, I observed various shop-floor operations including material handling, marking and cutting, fit-up, welding, machining, quality inspection, surface preparation, painting, and maintenance activities.

The training also enhanced my understanding of engineering drawings**, Bill of Materials** (BOM), **welding procedures** (Arc, MIG & TIG), **quality control techniques**, **safety practices** (PPE), and industrial documentation. Through interaction with engineers and supervisors, I developed a deeper understanding of production planning, process flow, teamwork, and the importance of maintaining quality and safety standards in heavy engineering projects.

This report summarizes my learning experience, practical observations, and the technical knowledge gained during the 30 days of vocational training, highlighting how the experience strengthened my engineering skills and industrial awareness.

➢ OVERVIEW

- Verification of raw materials using Purchase Orders and Material Test Certificates.
- Inspection of dimensions, thickness, grade, and surface defects.
- Material identification and traceability through heat numbers.

Material Receipt & Inspection

- Proper storage based on material type and size.
- Material issued according to production requirements.
- Traceability maintained throughout the process.

Material Storage & Issue

- Reading fabrication drawings and marking dimensions.
- Cutting using gas, plasma, or laser cutting methods.
- Edge preparation and grinding before fabrication.

Marking & Cutting

- Assembly of components using jigs and fixtures.
- Alignment, clamping, and tack welding operations.
- Checking dimensional accuracy and joint gaps.

Fabrication & Fit-Up

- Observation of Arc, MIG, and TIG welding processes.
- Understanding WPS and welding parameters.
- Learning about welding consumables and shielding gases.

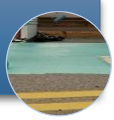
Welding Operations

- Visual inspection of weld joints and fabricated parts.
- Dimensional checking using Vernier and Micrometer.
- Awareness of NDT methods and quality documentation.

Quality Inspection

- Shot blasting and surface cleaning before coating.
- Application of primer and final paint coatings.
- Dry Film Thickness (DFT) measurement for quality.

Surface Preparation & Painting



- Final quality verification and documentation.
- Packing, marking, and loading of fabricated structures.
- Safe dispatch and delivery to project sites.

Final Inspection & Dispatch

➢ HEALTH & SAFETY INDUCTION

Tata Projects follows a strict **'Zero Harm'** Environment, Health, and Safety **(EHS)** policy, integrating safety standards into all daily operations, contracting, and management systems. The guidelines require 100% compliance with **Personal Protective Equipment (PPE)**, mandatory incident reporting, and safe work practices across all construction and engineering sites.

To achieve and enforce these standards, Tata Projects implements several core components:

Leadership & Culture: Safety is a prime responsibility at all levels. Senior leadership regularly conducts virtual insight sessions and site boot camps to promote a safety-first culture.
Digitalization: The company uses the **TQDigi'lytics** platform to monitor real-time safety metrics and the **Safety Risk Index (SRI)** to proactively address potential severe events.

Mandatory Site Guidelines:

PPE Compliance: Hard hats, safety shoes, and high-visibility vests are mandatory at all times.
Housekeeping & Waste: Strict good housekeeping and waste management rules are enforced.
Permit to Work: High-risk tasks (e.g., working at heights, excavations) strictly require a prior Permit to Work System.
Designated Zones: Smoking, eating, and resting are restricted to designated, authorized areas.

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➢ DELIVERABLES PREPARED

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1. Initial Stage: Understanding Purchase Orders (POs)

- I began by carefully **reading** through the **Purchase Orders** to understand their structure and content.
- Each PO was then **summarized** to capture the essential details such as item descriptions, pricing, and quantities.
- This step helped me build a strong foundation for the subsequent analysis.

2. Comparative Study: PO vs Base Price

- After summarizing, I **compared** the prices listed in the **POs against the base prices.**
- This comparison allowed me to identify variations, highlighting where costs were aligned and where discrepancies existed.
- It provided the first layer of insight into **potential cost differences.**

3. Report Analysis of Cost Differences

- I prepared a detailed **report analysing the cost differences** observed in the comparison.
- The report included item-wise breakdowns, section-wise variations, and overall trends.
- This analysis was crucial in understanding the financial impact and areas requiring attention.

4. Data Visualization

- To make the findings clearer and more engaging, I created **visualizations such as charts and graphs.**
- These visuals illustrated cost differences across items and sections, making complex data easier to interpret.
- Visualization added value by enabling quick decision-making and better communication of results.

5. Quantity Verification

- Finally, I **reviewed the quantities** of all sections in each PO.
- This step ensured accuracy and consistency, confirming that the data matched the actual requirements.

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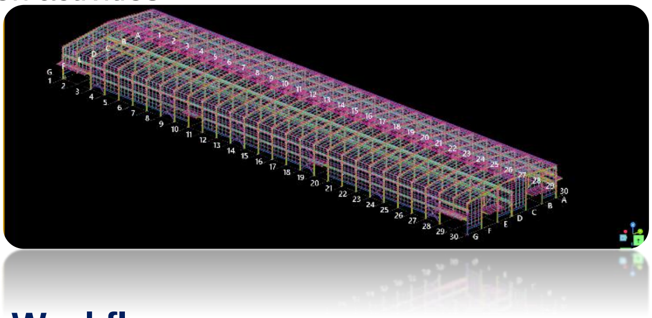
➢ DESIGN & DETAILING

The **Engineering Department** follows a systematic workflow to ensure design accuracy and smooth production planning.

- **Design & Approval Process**
  - Structural designs **(Stad-Pro, E-Tab)** are prepared using engineering standards and project specifications.
  - General Arrangement (**GA**) drawings are developed and reviewed in **AutoCAD** for approval.
  - After approval, the 3D model is finalized and released for detailing and production planning.

- **Detailing Using Tekla Structures**
  - Detailed fabrication models are prepared using **Tekla Structures** software.
  - Shop drawings are generated for fabrication activities.
  - The detailing stage also produces:
  - Assembly Drawings
  - Erection Drawings
  - **NC** (Numerical Control) Files
  - **DXF** & NC for **CNC cutting**
  - Bill of Materials (**BOM**)
  - **IFC** (Issued for Construction) drawings



- **Document Management & Project Workflow**
  - Engineering drawings are received and managed through the project document control system.
  - Drawings pass through stages such as **Received → Review → Stamping → Release → Publish** for use.
  - Approved drawings are finally released to the planning and fabrication departments for execution.

✓ Key Learning

This exposure helped me understand the complete engineering documentation cycle, the role of **AutoCAD** and **Tekla Structures**, and how digital models are converted into fabrication-ready drawings and production documents.

➢ SHOP VISITS

I visited different sections of the Structural Fabrication Shop and observed the complete manufacturing process from fabrication to surface finishing. These visits helped me understand industrial practices, equipment operation, and quality standards followed in heavy steel fabrication.

- **Surface Preparation by Abrasive Blasting**

  - Before painting, the steel surface is cleaned using abrasive blasting (**shot blasting**/grit blasting) to remove rust, mill scale, dust, and contaminants.
  - During blasting, **copper abrasive particles** (approximately **1-3 mm** in size) strike the steel surface at high velocity, creating a rough surface profile or microscopic **pits** (anchor pattern).
  - These tiny surface irregularities provide better mechanical adhesion for the **primer and paint coating**, resulting in stronger bonding, improved corrosion resistance, and a longer service life of the fabricated structure.

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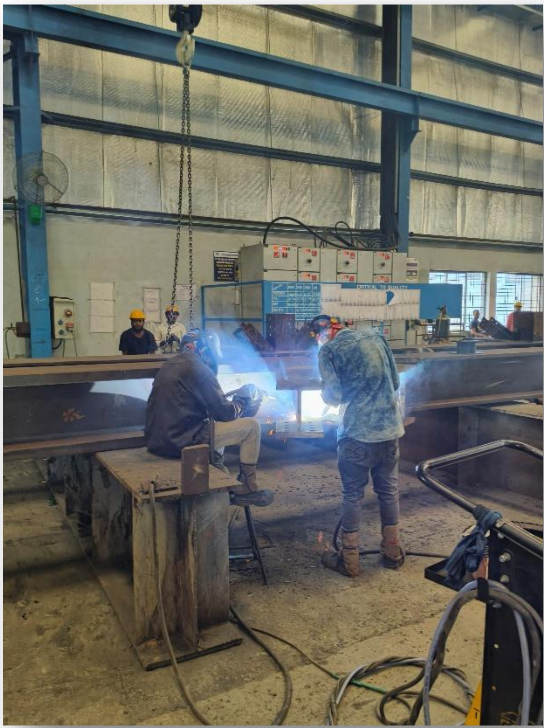
➢ SHOP VISITS

▪ Fabrication & Welding Shop



- Observed fit-up and assembly of structural members using beams, plates, and stiffeners.
- Studied **SMAW/MIG** welding operations and different weld joints used in fabrication.
- Learned the importance of **WPS** compliance, dimensional accuracy, and weld quality.

▪ Surface Preparation & Painting

- Understood surface cleaning by grinding and shot blasting before painting.
- Observed primer and final coating application using **airless spray painting equipment**.
- Learned about paint types (**Epoxy, PU** and **Phenolic**) and corrosion protection methods, and the significance of **WFT (Wet Film Thickness)** measurement for ensuring proper coating performance.

➢ SHOP VISITS

▪ Painting Equipment

- Studied the working principle of **Pot Gun & Airless Spray**, where high pressure atomizes paint without mixing with air, resulting in uniform coating and reduced paint wastage.
- Observed paint circulation, pressure control, and **spray gun** operation during coating activities.

▪ Quality Control & Inspection

- Learned about **Dry Film Thickness (DFT)** measurement and coating adhesion tests.
- Observed visual inspection and documentation before quality clearance.
- Understood the importance of **QA/QC** records for final acceptance.

➢ SHOP VISITS

▪ Final Inspection & Dispatch

- Witnessed final visual inspection, identification tagging, and documentation verification.
- Observed packing and loading operations using **cranes and forklifts**.
- Learned about dispatch documents such as Dispatch Note, Invoice,

✓ Key Learning

The shop visits provided practical exposure to **Industrial Fabrication**, **Welding**, **Painting**, **Quality assurance, Quality control** and **Dispatch Activities**, helping me connect theoretical engineering concepts with real-time manufacturing practices.

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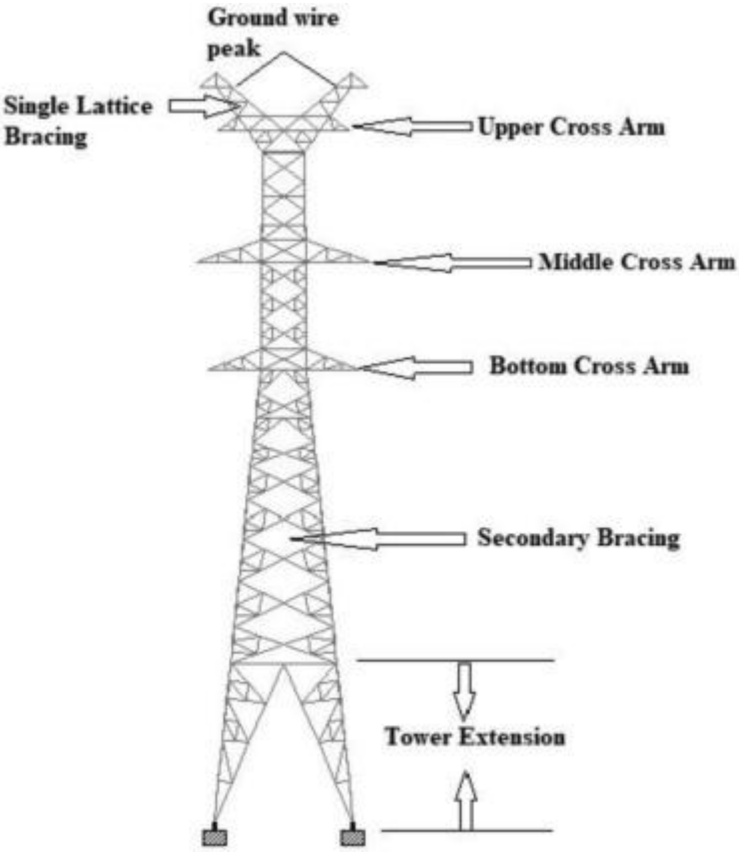
➢ PROTO YARD VISIT

Transmission towers are primarily classified as **suspension** or **tension** structures based on their mechanical function, how they handle conductor loads, and the angle of the transmission line.



1. Suspension Towers (Tangent Towers)

Suspension towers are the most common type used in electrical grids. They are designed for straight, uninterrupted sections of a transmission line.

Function: They primarily bear the **downward weight** (gravity) of the conductors and insulators and the **lateral wind load**.
Tension: The mechanical tension on either side of the tower remains equal, meaning these towers do not carry longitudinal (forward/backward) pulling forces.
Sub-types: These are typically designated as **Type A** towers, which allow for a very minor line deviation (usually less than 2°)

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➢ PROTO YARD VISIT

Suspension tower

Tension tower

Transposition tower

2. Tension Towers (Angle or Strain Towers)

Tension towers are built much **more robustly** than suspension towers. They are installed at locations where the transmission line is subjected to **high axial loading or strain**.

Function: They are utilized when the power line changes direction (angles), terminates, or requires extra-long spans such as crossing rivers, railways, or valleys.
Tension: These towers are designed to handle immense **horizontal pulling forces**. The cables pull significantly on the structure from the direction of the line.
Sub-types: They are categorized based on the degree of the angle deviation in the line:
Type B: Small angle deviation (2° to 15°).
Type C: Medium angle deviation (15° to 30°).
Type D: Large angle deviation (30° to 60°).
Type E (Dead-End): Maximum tension, used for the exact termination or starting of a line.

➢ QUALITY INSPECTION

✦ Raw Material (RM) – HT & MS

- **HT (High-Tensile) Steel** and **MS (Mild Steel)** are inspected for **yield strength** and **ultimate tensile strength (UTS)**.
- **Yield Test:** Determines the stress at which material begins to deform plastically.
- **UTS Test:** Measures the maximum stress material can withstand before failure.
- Results ensure conformity with design and safety standards before production.

✦ Inspection Steps during Galvanisation

- **Surface Preparation:** Cleaning, degreasing, and pickling to remove oxides and impurities.
- **Fluxing:** Ensures proper adhesion of zinc coating.
- **Galvanising Bath:** Monitoring bath temperature and zinc purity.
- **Post-Treatment:** Checking coating thickness, uniformity, and visual defects (runs, bare spots).
- **Final Inspection:** Adhesion test and coating weight verification.

✦ Non-Destructive Testing (NDT)

- Used to detect internal or surface defects **without damaging components**.
- Common methods:
    - **Ultrasonic Testing (UT):** Detects internal flaws and thickness variations.
    - **Magnetic Particle Testing (MT):** Reveals surface cracks in ferromagnetic materials.
    - **Dye Penetrant Testing (PT):** Highlights surface discontinuities on non-porous materials.

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Thank You!